

Haoyu Chen

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Education

2022–2026 **The Hong Kong University of Science and Technology (Guangzhou)**

Ph.D. in Robotics and Autonomous Systems

Advisors: Prof. Lei Zhu and Prof. Ping Tan

2017–2021 **The Chinese University of Hong Kong, Shenzhen**

B.ENG. in Computer Science and Engineering

Advisor: Prof. Xiaoguang Han

Research Experience

2026 **Tongyi Lab**, *Hangzhou, China*

- Research Scientist
- Core contributor to Wan Video 2.7 & 3.0's post-training stage.
- I focused on enhancing the model's visual quality, realism, and aesthetic appeal.

2025 **Adobe**, *San Jose, US*

- Research Intern, Supervisor: Zhe Lin and Qing Liu
- A Unified Autoregressive Multimodal Framework for Interleaved Image-Text Generation

2021–2022 **Xiaobing.AI**, *Shanghai, China*

- Research Intern, Supervisor: Baoyuan Wang
- Audio-driven Talking Face Generation

2020 **Amazon Web Services (AWS) Shanghai AI Lab**, *Shanghai, China*

- Research Intern, Supervisor: Zhi Zhang
- Image and Video Super-Resolution

Research Interests

I am currently focused on **video generation**, especially post-training for enhancing the visual quality, realism, and aesthetic appeal of generative video models. My research background includes **low-level vision**, particularly **image and video processing and restoration**, as well as **generative models**, **multi-modal large language models**, and **agent-based systems**.

Publications

First Author Publications

arXiv 2026 **How Long Can Unified Multimodal Models Generate Images Reliably? Taming Long-Horizon Interleaved Image Generation via Context Curation**

Haoyu Chen, Qing Liu, Yuqian Zhou, He Zhang, Zhaowen Wang, Mengwei Ren, Jingjing Ren, Xiang Wang, Zhe Lin, Lei Zhu

CVPR 2025 **POSTA: A Go-to Framework for Customized Artistic Poster Generation**

Haoyu Chen*, Xiaojie Xu*, Wenbo Li, Jingjing Ren, Tian Ye, Songhua Liu, Ying-Cong Chen, Lei Zhu, Xinchao Wang

- CVPR 2026 **PhotoArtAgent: Intelligent Photo Retouching with Language Model-based Findings Artist Agents**
Haoyu Chen, Keda Tao, Yizao Wang, Xinlei Wang, Lei Zhu, Jinjin Gu
- CVPR 2025 **JarvisIR: Elevating Autonomous Driving Perception with Intelligent Image Restoration**
Yunlong Lin*, Zixu Lin*, Haoyu Chen*, Panwang Pan*, Chenxin Li, Sixiang Chen, Kairun Wen, Yeying Jin, Wenbo Li, Xinghao Ding
- NeurIPS 2024 **RestoreAgent: Autonomous Image Restoration Agent via Multimodal Large Language Models**
Haoyu Chen, Wenbo Li, Jinjin Gu, Jingjing Ren, Sixiang Chen, Tian Ye, Renjing Pei, Kaiwen Zhou, Fenglong Song, Lei Zhu
- CVPR 2024 **Low-Res Leads the Way: Improving Generalization for Super-Resolution by Self-Supervised Learning**
Haoyu Chen, Wenbo Li, Jinjin Gu, Jingjing Ren, Haoze Sun, Xueyi Zou, Youliang Yan, Zhensong Zhang, Lei Zhu
- IEEE Transactions on Multimedia **Density-Aware Video Desnowing with Robust Alignment on a Large-Scale Dataset**
Haoyu Chen, Jingjing Ren, Jiaying Shen, Sixiang Chen, Jinjin Gu, Ping Tan, and Lei Zhu
- ICCV 2023 **Snow Removal in Video: A New Dataset and A Novel Method**
Haoyu Chen, Jingjing Ren, Jinjin Gu, Hongtao Wu, Xuequan Lu, Haoming Cai, Lei Zhu
- CVPR 2023 **Masked Image Training for Generalizable Deep Image Denoising**
Haoyu Chen*, Jinjin Gu*, Yihao Liu, Salma Abdel Magid, Chao Dong, Qiong Wang, Hanspeter Pfister, Lei Zhu
- arXiv 2021 **Attention in attention network for image super-resolution**
Haoyu Chen*, Jinjin Gu, Zhi Zhang

Selected Co-author Publications

- ICCV 2025 **Turbo2K: Towards Ultra-Efficient and High-Quality 2K Video Synthesis**
Jingjing Ren, Wenbo Li, Zhongdao Wang, Haoze Sun, Bangzhen Liu, Haoyu Chen, Jiaqi Xu, Aoxue Li, Shifeng Zhang, Bin Shao, Yong Guo, Lei Zhu
- NeurIPS 2025 **JarvisArt: Liberating Human Artistic Creativity via an Intelligent Photo Retouching Agent**
Yunlong Lin, ZiXu Lin, Kunjie Lin, Jinbin Bai, Panwang Pan, Chenxin Li, Haoyu Chen, Zhongdao Wang, Xinghao Ding, Wenbo Li, Shuicheng YAN
- NeurIPS 2025 **CamEdit: Continuous Camera Parameter Control for Photorealistic Image Editing**
Xinran Qin, Zhixin Wang, Fan Li, Haoyu Chen, Renjing Pei, Wenbo Li, Xiaochun Cao
- NeurIPS 2024 **UltraPixel: Advancing Ultra-High-Resolution Image Synthesis to New Peaks**
Jingjing Ren*, Wenbo Li*, Haoyu Chen, Renjing Pei, Bin Shao, Yong Guo, Long Peng, Lei Zhu
- CVPR 2024 **CoSeR: Bridging Image and Language for Cognitive Super-Resolution**
Haoze Sun, Wenbo Li, Jianzhuang Liu, Haoyu Chen, Renjing Pei, Xueyi Zou, Youliang Yan, Yujiu Yang
- IJCV 2024 **Triplane-Smoothed Video Dehazing with CLIP-Enhanced Generalization**
Jingjing Ren, Haoyu Chen, Tian Ye, Hongtao Wu, Lei Zhu

- arXiv 2024 **Towards Flexible, Scalable, and Adaptive Multi-Modal Conditioned Face Synthesis**
Jingjing Ren, Cheng Xu, **Haoyu Chen**, Xinran Qin, Chongyi Li, Lei Zhu
- ECCV 2020 **PIPAL: a Large-Scale Image Quality Assessment Dataset for Perceptual Image Restoration**
Jinjin Gu, Haoming Cai, **Haoyu Chen**, Xiaoxin Ye, Jimmy S. Ren, Chao Dong

Awards & Competitions

- 2018 **Winner (1st Place)**, *IEEE ICDM 2018 Global AI Challenge on Meteorology*
2020 **4th Place**, *ECCV 2020 Workshop AIM 2020: Video Extreme Super-Resolution*

Research Service

Conference Reviewer

Conference on Computer Vision and Pattern Recognition (CVPR), 2024, 2025
European Conference on Computer Vision (ECCV), 2024
The Neural Information Processing Systems (NeurIPS), 2024, 2025
The International Conference on Learning Representations (ICLR), 2025
International Conference on Artificial Intelligence and Statistics (AISTATS), 2025
ACM Multimedia (ACM MM), 2023, 2024
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2023
SIAM International Conference on Data Mining (SDM), 2024
Winter Conference on Applications of Computer Vision (WACV), 2023, 2024

Journal Reviewer

IEEE Transactions on Multimedia (TMM)
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
IEEE Transactions on Mobile Computing (TMC)